

Question_2:

Code:

```
1  /*
2  Question_2:
3  Sort the given array using Selection Sort.
4  */
5
6  #include <iostream>
7  using namespace std;
8
9  void swap(int *x, int *y) {
10     int t = *x;
11     *x = *y;
12     *y = t;
13 }
14 void selectionSort(int arr[], int n) {
15     for (int i = 0; i < n - 1; i++) {
16         int m = i;
17
18         for (int j = i + 1; j < n; j++)
19             if (arr[j] < arr[m])
20                 m = j;
21         swap(&arr[i], &arr[m]);
22     }
23 }
24 void printArray(int arr[], int n) {
25     for (int i = 0; i < n; i++)
26         cout << arr[i] << " ";
27 }
28 int main() {
29     int n;
30     cin >> n;
31
32     int arr[n];
33     for (int i = 0; i < n; i++)
34         cin >> arr[i];
35     selectionSort(arr, n);
36     printArray(arr, n);
37     return 0;
38 }
```

Output:

Test Case_1:

```
5
15 17 11 25 1
1 11 15 17 25
PS C:\Users\Prabin Yadav
```

Test Case_2:

```
6
65 45 27 21 25 11
11 21 25 27 45 65
PS C:\Users\Prabin Yadav
```